Goal: Be able to solve (very basic) differential equations by integration.

Definition. For $y = f(x)$, a **differential equation** is an equation involving x, y, and the derivatives of y.

Here are some examples (m and k are constants):

Equation	English Translation
$\frac{d^2y}{dx^2} = -9.81$	"Acceleration due to gravity is constant."
$\frac{dy}{dx} = ky$	"The rate a population grows is proportional to the amount of organisms in the population."
$\frac{d[y]}{dt} = -k[x_{(t)}][y_{(t)}]$	"The speed of a chemical reaction is proportional to the number of interactions between the molecules."
$m\frac{d^2y}{dx^2} + k\sin y = 0$	"The net force acting on an oscillating pendulum is 0."

$2x+1=5$

Example 1. Verify $y = 3e^{2x}$ is a solution to the differential equation $y' = 2y$.

LHS: $y' = [3e^{2x}]' = 3e^{2x}\cdot 2 = 6e^{2x}$

RHS: $2y = 2(3e^{2x}) = 6e^{2x}$

Example 2. Solve the differential equation $y' = \frac{1}{x}$.

$2x+1=5$

$2x=4$

$x=2$

$y' = \frac{1}{x}$

$\int y'\,dx = \int \frac{1}{x}\,dx$

$y_{(x)} = \ln|x| + C$

Definition. The **general solution** to a differential equation is a function that describes all possible solutions of the differential equation.

Example 3. Find the general solution to $y' = 3x^2$

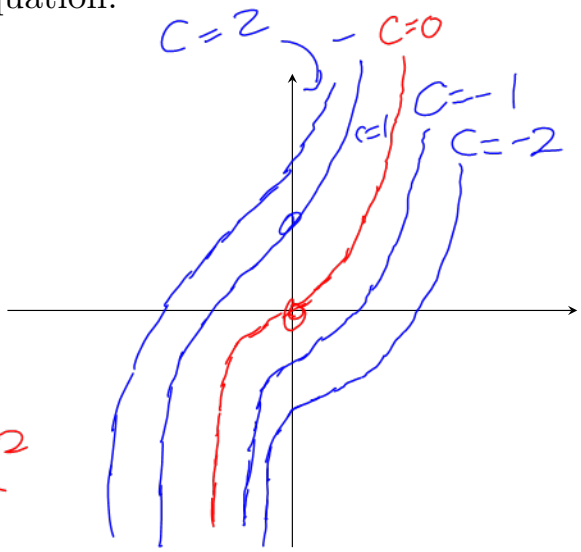
$$y' = 3x^2$$

$$\int y'\,dx = \int 3x^2\,dx$$

$$y(x) = 3\frac{x^{2+1}}{2+1} + C$$

$$y(x) = x^3 + C$$

The general solution gives us a whole family of solutions to our differential equation.



$y' = 3x^2$

Question: Is there a way to have a unique (i.e., only 1) solution?

Answer: Ask that our solution passes through a point (e.g., $y(0) = 0$)

Definition. The extra condition above is an example of an initial condition. Usually, we specify when $x = 0$ (but that is not required).

IVPs

Definition. An **initial value problem (IVP)** is a differential equation combined with initial conditions.

Example 4. The system

$$\begin{cases} y' = \sqrt{x} \\ y(0) = 1 \end{cases}$$

is an IVP. How can we solve it?

Step 1 Find the general solution

$y' = \sqrt{x}$

$\int y' dx = \int x^{\frac{1}{2}} dx$

$y = \frac{x^{\frac{1}{2}+1}}{\frac{1}{2}+1} + C$

$y = \frac{2}{3} x^{\frac{3}{2}} + C$

Step 2 Use the IC to solve for C

$y(0) = 1$; I should

$y(0) = \frac{2}{3} \cdot 0^{\frac{3}{2}} + C$

$1 = C$

Conclusion: The solution to the IVP is

$y(x) = \frac{2}{3} x^{\frac{3}{2}} + 1$

$y = kx$ Proportionality Constant

Definition. We say y is **(directly) proportional to** x if there is a constant k such that $y = kx$.

Example 5. The size of a population is governed by

$$\frac{dP}{dt} = \sqrt[3]{t}$$

where P is the population after t days (i.e., $\frac{dP}{dt}$ is proportional to $\sqrt[3]{t}$ with a constant coefficient of 1). If the initial population is 1000, find the population size after 8 days.

$\frac{dP}{dt}$ is proportional to $\sqrt[3]{t}$ where $k=1$

Need to solve

$$\begin{cases} \frac{dP}{dt} = \sqrt[3]{t} \\ P(0) = 1000 \end{cases}$$

$$\frac{dP}{dt} = \sqrt[3]{t} \qquad t^{\frac{1}{3}}$$

$$\int \frac{dP}{dt}\,dt = \int \sqrt[3]{t}\,dt$$

$$P(t) = \frac{t^{\frac{1}{3}+1}}{\frac{1}{3}+1} + C$$

$$P(t) = \frac{3}{4} t^{\frac{4}{3}} + C$$

$P(0) = 1000$: 1000 should $P(0) = 0 + C = C$

$C = 1000$

$$\Rightarrow \boxed{P(t) = \frac{3}{4} t^{\frac{4}{3}} + 1000}$$

$$P(8) = \frac{3}{4}\left(\sqrt[3]{8}\right)^4 + 1000 = \frac{3}{4} \cdot 16 + 1000$$

$$= 12 + 1000 = 1012 \text{ people}$$

Multiple ICs

Fact: The have a unique solution, you need N ICs where N is the highest order of derivatives in the differential equation.

Example 6. Solve the IVP:

$$\begin{cases} y''(t) = t - 3 \\ y'(1) = \frac{1}{2} \\ y(0) = 1 \end{cases}$$



$$y''(t) = t - 3$$

$$\int y''(t)\,dt = \int (t-3)\,dt$$

$$y'(t) = \frac{1}{2}t^2 - 3t + C$$

$y'(1) = \frac{1}{2}$: should

$$y'(1) = \frac{1}{2} - 3 + C$$

$$\frac{1}{2} = \frac{1}{2} - 3 + C$$

$$0 = -3 + C$$

$$C = 3$$

$$y'(t) = \frac{1}{2}t^2 - 3t + 3$$

$\Rightarrow$

$$\int y'(t)\,dt = \int \left(\frac{1}{2}t^2 - 3t + 3\right) dt$$

$$y(t) = \frac{1}{2}\cdot\frac{1}{3}t^3 - \frac{3}{2}t^2 + 3t + D$$

$y(0) = 1$: should

$$y(0) = 0 - 0 + 0 + D$$

$$D = 1$$

$\Rightarrow$

$$y(t) = \frac{1}{6}t^3 - \frac{3}{2}t^2 + 3t + 1$$

Example 7. A ball is kicked off the ground with an initial velocity of $20\,\mathrm{m\,s^{-1}}$. Describe the height of the ball as a function of time (assume gravity is the only force acting on the ball once it is kicked).

$20\frac{m}{s}$

$32\frac{ft}{s^2}$

Need to solve s'' ~~$a(t) = -9.81$~~

$s'(0)$ ~~$v(0) = 20$~~

$s(0) \simeq 0$

Position $\xrightarrow{d/dt}$ Velocity $\xrightarrow{d/dt}$ Acceleration

Position $\xleftarrow{\int}$ Velocity $\xleftarrow{\int}$ Acceleration

$a(t) = -9.81$

$v(t) = -9.81t + C$

$v(0) = 20$

$20 = v(0) = C$

$\Rightarrow \quad v(t) = -9.81t + 20$

$s(t) = -\frac{9.81}{2}t^2 + 20t + D$

$s(0) = 0 \Rightarrow D = 0$

$s(t) = -4.905t^2 + 20t$

Example 8 (HW 27, Q6)**.** Solve the IVP:

$$\begin{cases} y''(x) = 4\sin x + \frac{10}{\pi^2}x \\ y'(\pi) = 4 \\ y\left(\frac{\pi}{2}\right) = -4 \end{cases}$$

$$y''(x) = 4\sin x + \frac{10}{\pi^2}x$$

$$\int y''(x)\,dx = \int \left(4\sin x + \frac{10}{\pi^2}x\right)dx$$

$$y'(x) = 4(-\cos x) + \frac{10}{\pi^2}\cdot\frac{1}{2}x^2 + C$$

$$y'(x) = -4\cos x + \frac{5}{\pi^2}x^2 + C$$

$y'(\pi) = 4$ Should $y'(\pi) = -4\cos(\pi) + \frac{5}{\pi^2}(\pi)^2 + C$

$$4 = 4 + 5 + C$$

$$0 = 5 + C$$

$$C = -5$$

$$y'(x) = -4\cos x + \frac{5}{\pi^2}x^2 - 5$$

$$\int y'(x)\,dx = \int \left(-4\cos x + \frac{5}{\pi^2}x^2 - 5\right)dx$$

$$y(x) = -4\sin x + \frac{5}{\pi^2}\cdot\frac{1}{3}x^3 - 5x + D$$

$y\left(\frac{\pi}{2}\right) = -4$ Should $y\left(\frac{\pi}{2}\right) = -4\sin\left(\frac{\pi}{2}\right) + \frac{5}{3\pi^2}\left(\frac{\pi}{2}\right)^3 - 5\left(\frac{\pi}{2}\right) + D$

$$-4 = -4 + \frac{5\pi}{24} - \frac{5\pi}{2} + D$$

$$D = \frac{5}{2}\pi - \frac{5}{24}\pi$$

$$D = \frac{55}{24}\pi$$

$$\Rightarrow \quad y(x) = -4\sin x + \frac{5}{3\pi^2}x^3 - 5x + \frac{55}{24}\pi$$